

Deployment Verification Report

Target network: Base Sepolia

Local verification network: Anvil chain ID 31337

Deployment toolchain: Foundry scripts only

Status: Local Anvil deployment verified. Base Sepolia deployment waits for private RPC, deployer key, and explorer API key.

1. Verification Purpose

This report documents the post-deployment checks that prove the deployed protocol matches the course requirements. The goal is to catch deployment mistakes that a normal compile or unit test cannot detect: wrong Timelock delay, wrong Governor parameters, forgotten deployer admin rights, missing treasury ownership, or incorrect proxy address wiring.

2. Scripts

Script	Responsibility
<code>script/Deploy.s.sol</code>	Deploys the full protocol, performs role handoff, delegates initial votes, and writes manifest JSON.
<code>script/VerifyDeployment.s.sol</code>	Reads deployment manifest and validates post-deployment role and governance invariants.
<code>script/UpgradeTBillToken.s.sol</code>	Executes the V1 to V2 upgrade path for the UUPS token.
<code>script/GasCompare.s.sol</code>	Measures representative protocol operation gas and writes the gas report.

3. Required Checks

Check	Expected value	Why it matters
Timelock delay	2 days	Gives users and reviewers a window before privileged execution.
Governor voting delay	1 day	Prevents instant proposal voting.
Governor voting period	1 week	Gives voters time to react.
Governor quorum	4%	Matches project requirement.
Proposal threshold	1% of governance token supply	Reduces proposal spam.
TBILL admin/upgrader	Timelock	Prevents deployer-controlled minting, pausing, or upgrades after setup.
Vault admin	Timelock	Keeps yield and pause powers governance-controlled.
Treasury manager	Timelock	Ensures treasury actions go through governance.
Timelock proposer	Governor	Only successful proposals can be queued.
Timelock executor	Open executor <code>address(0)</code>	Anyone can execute once the Timelock delay has elapsed.

4. Local Anvil Verification Output

```
Deployment verification passed for chain 31337
Timelock 0x0165878A594ca255338adfa4d48449f69242Eb8F
Governor 0xa513E6E4b8f2a923D98304ec87F64353C4D5C853
TBill proxy 0xDc64a140Aa3E981100a9becA4E685f962f0cF6C9
```

5. Base Sepolia Deployment Inputs

Real Base Sepolia deployment is pending local secret configuration:

- `BASE_SEPOLIA_RPC_URL`
- `PRIVATE_KEY`
- `BASESCAN_API_KEY`
- funded deployer account
- optional `WALLETCONNECT_PROJECT_ID` for frontend deployment
- optional `SUBGRAPH_DEPLOY_KEY` for Subgraph Studio deployment

6. Final Submission Links

Paste verified Base Sepolia explorer links here after running:

```
npm run deploy:base-sepolia  
npm run verify:base-sepolia
```

Contract	Base Sepolia address	Explorer verification link
TBillToken proxy	TBD	TBD
TBillToken implementation	TBD	TBD
RwaGovernanceToken	TBD	TBD
ReserveCertificateNFT	TBD	TBD
TBillVault	TBD	TBD
RwaStableAMM	TBD	TBD
OracleAdapter	TBD	TBD
ProtocolTreasury	TBD	TBD
ProtocolFactory	TBD	TBD
RwaGovernor	TBD	TBD
TimelockController	TBD	TBD